Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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ABSTRACT

Stablecoins are designed to trade near one dollar, but stress events repeatedly create visible deviations across venues, quote currencies, and settlement rails. These deviations often appear to be arbitrage opportunities. In practice, however, a quoted price gap is not the same as an executable trade: a trader must clear order-book depth, absorb VWAP slippage, pay fees, and bear settlement risk before any apparent spread becomes profit.

We introduce Stablecoin StressBench, an execution-aware benchmark for evaluating stablecoin dislocation models. The benchmark separates three layers: optical price dislocations, executable arbitrage opportunities, and ex-ante predictable executable opportunities. During the March 2023 USDC/SVB stress window, 35.1% of 1-minute windows exceed a 10 bps primary/max cross-quote basis threshold on price alone, while only 2.88% remain profitable at $10K notional after VWAP book-walking and trading costs. A hindsight oracle earns positive net bps, confirming that profitable windows exist, but rule-based models, ML classifiers, and a direct expected-net-profit regressor all lose money out of sample.

The result reframes stablecoin arbitrage prediction as an execution-barrier problem rather than a price-signal detection problem. Stablecoin StressBench contributes a historical stress-event catalogue, execution-aware labels, economic evaluation metrics, and a model ladder for future work on crypto market microstructure, optimal execution, and financial benchmark construction.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 INTRODUCTION

Stablecoins are designed to make a simple promise: one token should trade close to one dollar. This promise makes stablecoins central to crypto-market settlement, quoting, collateral, and liquidity transfer. When confidence in that promise weakens, prices fragment across venues, quote currencies, and liquidity pools.

Price fragmentation often appears to create arbitrage. During a stress event, one route may imply that USDC is cheap while another implies that USDT or USD is expensive. A trader looking only at quoted prices may conclude that buying through one route and selling through another should generate mechanical profit.

That conclusion is incomplete. A quoted price gap is not a trade. To realize arbitrage, a trader must clear available order-book depth, absorb VWAP slippage, pay taker fees, and bear settlement delay. These frictions are small individually but large jointly. As a result, many stablecoin dislocations are optical rather than executable.

This paper introduces Stablecoin StressBench, an execution-aware benchmark for stablecoin dislocations. The benchmark separates three layers: optical price dislocations, executable arbitrage opportunities, and ex-ante predictable executable opportunities. The distinction matters because a model can classify price dislocations accurately while still losing money if the selected windows do not survive execution.

We study this distinction during the March 2023 USDC/SVB stress event and in a broader historical catalogue of stablecoin stress episodes. In the primary Tier-A execution-grade test window, 35.1% of minutes exceed a 10 bps primary/max cross-quote basis threshold, but only 2.88% remain profitable at $10K notional after VWAP execution and costs. A hindsight oracle confirms that profitable windows exist, yet rule-based models, ML classifiers, and direct expected-profit regressors all fail to capture them out of sample. The result reframes stablecoin arbitrage prediction as an execution-barrier problem: the hard problem is not detecting that a stablecoin price moved, but identifying which dislocations survive execution.

Contributions. We make four contributions.

(1) Execution-aware stablecoin benchmark. Per-minute labels distinguish quoted price dislocations from executable arbitrage opportunities using order-book depth, VWAP book-walking, fees, notional size, and settlement penalties.
(2) Historical stress-event framework. We catalogue stablecoin stress events across mechanism classes and classify each event by data availability. Tier-A events support execution-aware labels; Tier-B events support price and liquidity analysis; Tier-C events support historical taxonomy only.
(3) Price-to-execution and oracle-gap evidence. In the USDC/SVB Tier-A test event, 35.1% of minutes exceed a 10 bps primary/max cross-quote basis threshold, but only 2.88% remain executable at $10K. A hindsight oracle is profitable, while all tested deployable models are economically negative (oracle gap $>$ 210 bps).
(4) Literature-motivated model ladder. We evaluate economic anchors, naive price rules, time-series baselines, regularized linear models, tree ensembles, direct expected-profit

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regression, meta-labeling, and regime detectors. This ladder establishes reproducible baseline failures for future models attempting to close the oracle gap.

This paper fits ICAIF as financial benchmark construction for blockchain and cryptocurrency markets, combining market microstructure, trading and execution, validation and calibration of financial models, and economic evaluation of ML systems.

2 RELATED WORK

Stablecoin stress and depegs. The USDC/SVB episode (March 2023) produced well-documented price fragmentation across venues; we focus on the ex-ante predictability question that event documentation does not address. Kwon et al. [1] analyse Terra/LUNA collapse mechanics; we use May 2022 as our validation split event. Griffin and Shams [2] study Tether price manipulation; our contribution is execution microstructure, not flow manipulation.

Limits to arbitrage. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish the general theory. Hautsch et al. [3] model blockchain-specific limits to arbitrage (settlement finality, gas costs, latency); Makarov and Schoar [8] provide empirical evidence of persistent crypto price gaps across venues. We provide an execution-aware empirical measurement of stablecoin arbitrage barriers during a real stress event, using documented order-book depth provenance.

Market microstructure and optimal execution. Glosten and Milgrom [7] establish the theoretical basis for adverse selection during stress. Cont and Kukanov [6] develop optimal order-routing across fragmented limit-order books, which motivates our VWAP depth-walk formulation. Adams et al. [9] characterise AMM liquidity dynamics, informing our settlement-penalty modelling.

Financial ML benchmarks and model validation. López de Prado [10] introduces meta-labeling to separate primary signal generation from execution filtering directly applicable to our setting where price signals are plentiful but most are not executable. Cintra and Holloway [11] apply BOCPD to Curve StableSwap pool reserves and detect the May 2022 Terra/UST de-peg with a 12-hour lead; we find BOCPD achieves only AUROC 0.23 on CEX cross-quote data, suggesting the pool-reserve signal does not transfer directly. No prior benchmark provides execution-aware stablecoin arbitrage labels with real L2 depth provenance.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The gold-layer dataset contains 47,487 1-minute bars across three event windows (Table 1). Price data comes from the Binance Vision public archive (klines, aggTrades) and Coinbase REST candles. Execution-grade order-book depth comes from Binance USDM futures bookDepth snapshots and incremental updates. We track four instruments BTC-USDC, BTC-USDT, ETH-USDC, ETH-USDT across Binance, Coinbase, and Kraken.

Table 1: Dataset coverage by split.

Split	Event	Minutes	Pos. rate
Train	Calm control (3 windows)	20,125	0.00%
Validation	Terra/LUNA, May 2022	11,523	2.45%
Test	USDC/SVB, Mar 2023	15,839	2.88%
Total		47,487	

Positive rate = executable arb at \$10K, 5-min horizon.

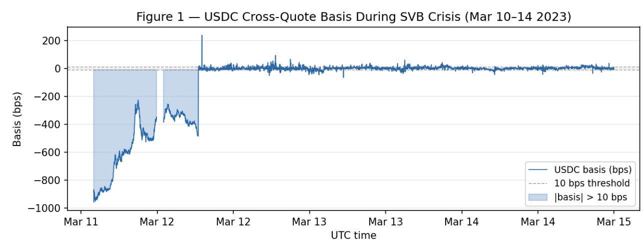

Figure 1: USDC-specific cross-quote basis (bps) during the SVB stress window (March 2023), exceeding 100 bps for sustained periods. The USDC-specific basis exceeds 10 bps in 12.65% of test minutes; the broader primary/max basis (max of USDC and USDT routes) exceeds 10 bps in 35.1%.

Depth provenance. Each row carries a `depth_source` tag: `real_l2_snapshot`, `real_l2_incremental`, or `synthetic_kline`. Net-profit labels use only `real_l2_*` rows; synthetic kline depth is isolated to a separate `is_paper_grade_depth` flag. Binance USDM futures bookDepth is the confirmed execution-grade source in the committed dataset. Full historical L2 for Coinbase and Kraken during the SVB period (March 2023) requires a Tardis subscription not included in the released benchmark.

3.2 Event-Based Split Design

Splits are defined by event identity, not by time alone (Figure 1). The train split contains three calm control windows (no stress event) that bracket the test period; the validation split uses the Terra/LUNA collapse (May 2022), a qualitatively similar but independent stress event; the test split uses the USDC/SVB de-peg (March 10–15, 2023).

This design has two guarantees: (i) no model trained on the train split has seen stress dynamics; (ii) threshold calibration on validation uses an independent stress event, not a held-out portion of the same episode. No lookahead: labels are constructed by `join_asof` at $t + h$ and formally verified. The Terra/LUNA validation split is not used to claim a second execution-grade test result; its role is calibration, exposing the thresholding procedure to independent stress dynamics before final evaluation on USDC/SVB.

Table 2: Data tier and permitted claims.

Tier	Data	Permitted claims
A (N=2)	Real L2 + trades	Execution gap, oracle gap, P&L, model evaluation
B (N=11)	OHLCV / pools	Peg magnitude (est.), liquidity stress
C (N=5)	Partial	Mechanism taxonomy and historical context only

Table 3: Stress-event roles in Stablecoin StressBench.

Event	Mechanism	Tier	Role
Terra/UST 2022	Algorithmic/reflexive	Validation	Threshold calibration
USDC/SVB 2023	Fiat-reserve bank shock	A	Main execution test
USDC recovery	Fiat-reserve recovery	A	Comparator
USDT/Curve 2023	DeFi pool imbalance	B	Price/liquidity context
FTX 2022	Exchange credit/liq.	B	Market stress context
BUSD 2023	Regulatory winddown	B	Issuer/regulatory context
IRON/TITAN 2021	Algorithmic/reflexive	C	Taxonomy context

3.3 Historical Stress Taxonomy and Data Tiering

Stablecoin stress is not a single phenomenon. We catalogue 18 stress events from 2020–2023 spanning seven distinct failure mechanisms: algorithmic/reflexive (FEI, IRON/TITAN, MIM, UST, USDD), fiat-reserve bank shock (USDC/SVB), regulatory/issuer winddown (BUSD, Binance conversion), exchange credit/liquidity (Celsius/3AC, HUSD, FTX [12]), DeFi pool imbalance (Curve/UST, USDT/Curve, USDC/DAI secondary), collateral liquidation (DAI Black Thursday), and RWA/niche stablecoin (Acala aUSD, USDR).

Events are classified by data availability tier, which governs what claims this paper may make (Table 2).
Execution-grade labels are available for the USDC/SVB stress window and its immediate post-stress recovery window (both Tier A). The price-to-execution gap, oracle net bps, and model evaluation in this paper are anchored to the USDC/SVB stress test; the recovery window serves as a Tier-A comparator, not an independent stress mechanism. Price-grade claims for Tier-B events are labelled as estimates (est.) and are excluded from any arbitrage-profitability argument.

3.4 Stress-Event Roles in the Benchmark

Table 3 clarifies how the historical catalogue is used. The catalogue establishes that stablecoin stress recurs across mechanisms, but only Tier-A events support execution-aware P&L claims. USDC/SVB is the primary execution-grade test; USDC recovery is a Tier-A comparator; and Tier-B / Tier-C events provide external validity and mechanism taxonomy rather than arbitrage-profitability evidence.

Terra/UST serves as the validation split (Tier B, Algorithmic/reflexive mechanism): it is used for threshold calibration and exposes the thresholding procedure to independent stress dynamics before final evaluation. It is not used to claim a second execution-grade result the validation split has executable-arb labels (3.08% positive at \$10K) but lacks the full Tier-A L2 depth reconstruction required for execution P&L inference.

Table 4: Three layers of stablecoin dislocation measured by Stablecoin StressBench (USDC/SVB test split, \$10K notional).

Layer	What it measures	SVB freq.
1. Optical dislocation	Quoted basis >10 bps on price data	35.1%
2. Executable opportunity	Net profit >0 after VWAP, fees, costs	2.88%
3. Predictable ex ante	Model identifies window before execution	No active model

Layer 3 frequency: every active non-oracle strategy that takes trades produces negative net bps; abstention (no-trade) is the zero-P&L floor.

3.5 Feature Sets

We define four nested feature sets from narrowest to broadest:

`price_only`	5 cross-quote basis columns
`price_plus_book`	+7 microstructure cols (spread, depth, imbalanc
`price_book_frag`	+3 cross-venue fragmentation cols
`price_book_settle`	+7 on-chain settlement proxies

The nesting structure isolates the marginal value of each information layer.

4 EXECUTION-AWARE LABELS

The benchmark measures three layers of stablecoin dislocation (Table 4). Understanding these layers clarifies why standard price-signal metrics are insufficient: a model can do well on Layer 1 while failing economically at Layer 2.
The gap between Layer 1 and Layer 2 is the price-to-execution gap; the gap between Layer 2 and Layer 3 is the oracle gap. Both are central measurements of this benchmark.

4.1 Cross-Quote Basis

The cross-quote USDC basis at time t is:

$$b_{\mathrm{USDC}}(t) = 10{,}000 \times \frac{P_{\text{BTC-via-USDC}}(t) - P_{\text{BTC-direct}}(t)}{P_{\text{BTC-direct}}(t)} \quad \text{[bps]}$$

where $P_{\text{BTC-via-USDC}}$ is the effective BTC price obtained by converting USD → USDC → BTC relative to the direct BTC-USD market. We also compute the max-absolute basis $b_{\max}(t) = \max(|b_{\mathrm{USDC}}|, |b_{\mathrm{USDT}}|)$ as a broader stress detector.

Basis labels for horizon $h \in \{1, 5, 15, 60\}$m and threshold τ:

$$y_{\mathrm{basis}}(t) = \mathbf{1}[\,|b_{\mathrm{USDC}}(t+h)| > \tau\,]$$

Primary classification task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution Walk

For each minute window and notional $q \in \{\$10\text{K}, \$50\text{K}, \$100\text{K}, \$500\text{K}\}$, we simulate a round-trip execution:

(1) Walk the available execution-grade bid and ask ladders for routes with confirmed real-L2 provenance to fill q notional, recording the VWAP price at each level consumed.
(2) Compute the gross spread: $\text{gross}(q) = \text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}$.
(3) Deduct taker fees on both legs and a settlement delay penalty.

4.3 Net Executable Arbitrage Labels

$$\text{net}(q, t) = \text{gross}(q, t) - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \tag{1}$$

$$y_{\text{arb}}(q, h, t) = \mathbf{1}\left[\max_{s \in [t, t+h]} \text{net}(q, s) > 0\right] \tag{2}$$

where $f_{\text{buy}}, f_{\text{sell}}$ are venue taker fees and δ_{settle} is a settlement penalty proxy. Primary economic task: `executable_arb_q10000_5m`. The forward-max in Eq. (2) asks: does a profitable window exist within the next h minutes? matching the practical constraint that a trader can execute at any point within the horizon.

4.4 Oracle Upper Bound

The oracle trades every minute where $\text{net}(q, t) > 0$ in hindsight. It uses information unavailable at prediction time and sets the theoretical performance ceiling. The oracle is not deployable; it exists to confirm that profitable windows exist and to calibrate the oracle gap.

5 MODELS AND EVALUATION

5.1 Model Ladder

The model stack is a ladder of increasingly realistic baselines, progressing from economic anchors through naive price rules, time-series persistence, interpretable linear models, nonlinear tabular models, direct expected-profit regression, and finance-specific filters. The purpose is not to exhaust all possible architectures; it is to establish whether progressively stronger and more economically aligned models can close the oracle gap.

Economic anchors. `NoTrade` (the economic floor: 0 net bps) and `Oracle` (hindsight ceiling).

Rule baselines. `PriceBasis10bps` (trade when $|b_{\text{USDC}}| >$ 10 bps), `PriceBasis25bps`, and `GrossArbThreshold` (gross spread > 0). These test whether threshold rules on price data are sufficient; motivated by [3], execution frictions are expected to make raw price rules uneconomic.

Statistical baselines. `LastValue` (AR0), `RollingMean`, `AR1` testing whether basis autocorrelation alone identifies executable stress. Each baseline uses an explicitly named lag column rather than assuming column ordering.

ML classifiers. Logistic (L2), Lasso (L1), Random Forest, XGBoost, and LightGBM each trained on the four nested feature sets. Tree ensembles are included for their strong tabular performance in financial time-series settings [8].

Expected-net-profit regressor. `ExpectedNetProfitRegressor` directly predicts $\text{net}(q, t)$ and trades when $\hat{y} > \theta^*$. If this also fails, the oracle gap is structural, not a label-mismatch artefact.

Table 5: Price-to-execution gap (test split, \$10K notional, 1-minute same-window horizon).

Thresh.	Price (max)	Price (USDC)	Exec. (\$10K)	Ratio
0 bps	97.15%	83.69%	3.34%	29×
5 bps	63.22%	25.77%	3.03%	21×
10 bps	35.09%	12.65%	2.88%	12×
25 bps	9.63%	6.53%	2.62%	3.7×

Meta-labeling filter. Motivated by [10]: the primary signal is $|b_{\text{USDC}}| > 10$ bps; a secondary LightGBM meta-model learns $\hat{g}_{\text{meta}} = \Pr[\text{net}(q, t) > 0 \mid \text{primary fires}]$ and filters predicted false positives.

Regime detectors. EWMA z-score, CUSUM, and BOCPD [11] identify stress regimes from the basis time series as two-stage precursors: detect stress first, then apply execution filters only during detected regimes.

5.2 Threshold Calibration

For probabilistic classifiers, the decision threshold is calibrated on the validation split:

$$\theta^* = \arg\max_{\theta \in [0.05, 0.95]} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i) \quad \text{s.t.} \quad |\{i : \hat{p}_i > \theta\}| \geq 25$$

The ≥ 25 trade minimum prevents degenerate abstention strategies from winning on small-sample artefacts.

5.3 Evaluation Protocol

Primary (economic): `net_bps_captured` (mean net bps per trade), `hit_rate_above_cost`, `final_pnl_usd`, `oracle_capture_pct` (= net bps / oracle net bps).

Secondary (statistical): AUROC, AUPRC, Brier score, n_{trades}.

The `NoTrade` baseline (= 0 net bps) is the economic floor: a model that loses money is worse than abstaining entirely, regardless of its AUROC.

6 RESULTS

6.1 Price Gaps Are Common, Execution Is Rare

Table 5 and Figure 2 show the central finding. At a 10 bps threshold, 35.09% of test-split minutes show a primary/max cross-quote basis exceeding the threshold on price alone; only 2.88% remain profitable after execution costs at \$10K notional a 12× price-to-execution ratio. The USDC-specific basis fires in 12.65% of minutes, reflecting that USDT routes also fragment during the SVB event. The gap persists at every threshold level, widening sharply below 10 bps as price signals fire constantly but order-book depth is thin.

Tracing the pipeline stage by stage: of 15,839 total test minutes, 5,560 (35.1%) show a max-basis price dislocation above 10 bps; 456 of those (2.88%) remain executable at \$10K/1m; the oracle trades 351 of those minutes profitably. Naive price rules trade far more often and lose money on

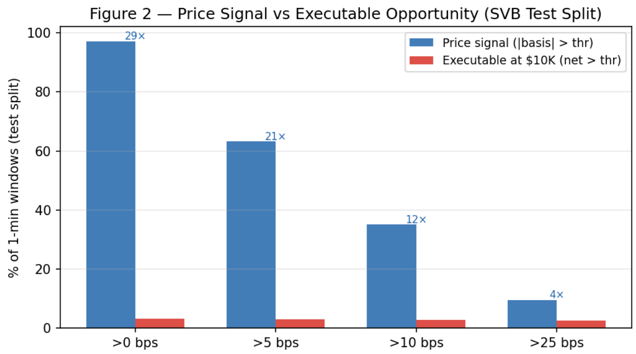

Figure 2: Price-to-execution gap by threshold. Price-signal frequency (dark) dwarfs executable frequency (light) at every level. The gap is largest at low thresholds, where price signals fire constantly but depth is thin.

Table 6: Oracle gap (test split). All non-oracle models are economically negative.

Task	Oracle	Best model	Gap
`basis_usdc_1m_gt10bps`	+161.7 bps	−49.1 bps	211 bps
`executable_arb_q10000_5m`	+224.6 bps	−42.9 bps	268 bps
`executable_arb_q50000_5m`	+146.8 bps	−112.7 bps	260 bps

Best model: logistic@price_plus_book (basis task);
price_threshold_25bps@price_only (arb tasks).

nearly all trades; calibrated ML models trade less aggressively or abstain entirely, but none achieves positive economic performance.

6.2 Profitable Windows Exist, But Only in Hindsight

Table 6 gives the oracle gap for each primary task. The oracle earns +161.7 bps on the `basis_usdc_1m_gt10bps` task. The best ML model (logistic, `price_plus_book` features) achieves −49.1 bps 211 bps below the oracle. On the executable arbitrage task, the oracle earns +224.6 bps; the best rule (`price_threshold_25bps`) achieves −42.9 bps. Every active non-oracle strategy that takes trades produces negative net bps on the test split; no-trade and meta-labeling (which abstains due to training-regime sparsity) remain at zero.

6.3 The Model Ladder Fails Economically

Table 7 summarises the model ladder. Price-threshold rules overtrade severely (611–2002 trades, −136 to −346 bps): the rules fire on price dislocations with no executable depth. Validation-thresholded ML classifiers trade less aggressively than naive rules but lower trade count does not translate into positive economics: the calibrated logistic model takes only 1 trade at −49.1 bps. LightGBM, despite higher AUROC (0.63), produces worse economic results (−329 bps) by taking 4 trades at large losses. Feature richness does not help:

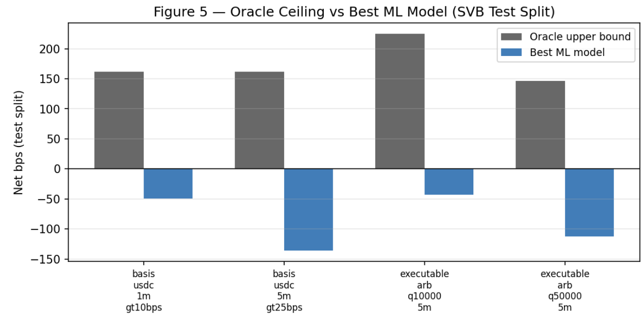

Figure 3: Oracle gap: grouped bars showing oracle net bps (gold) vs. best ML model (navy) for each task. The gap exceeds 200 bps across all tasks.

Table 7: Model ladder summary (task: **`basis_usdc_1m_gt10bps`**, test split). †Theoretical ceiling.

Model	Features	Net bps	Trades	AUROC
Oracle†		+161.7	316	
`logistic`	price_plus_book	−49.1	1	0.133
`lgbm`	price_only	−329.1	4	0.626
`gross_arb`	price_only	−136.0	611	0.530
`price_thr 10`	price_only	−269.5	2002	0.829
`price_thr 25`	price_only	−346.6	1033	0.743
`no_trade`		0.0	0	
`ExpNetProfit`	price_only	−61.4	1610	

the best logistic result uses `price_plus_book`; adding fragmentation or settlement features does not improve economic outcomes.

The `ExpectedNetProfitRegressor` (target: $\text{net}(q, t)$) achieves −61.4 bps a smaller loss than many rule baselines, but still economically negative. This is the strongest evidence that the oracle gap is structural: even when the training objective directly matches the economic evaluation criterion, the model cannot identify profitable windows out of sample.

6.4 Robustness and Diagnostics

Cost-regime robustness. The price-to-execution gap persists across all tested cost assumptions (Table 8). At base fees, 5.64% of windows are executable at \$10K/5m/10 bps; high fees reduce this to 5.46%. Adding a 10 bps settlement penalty gives 4.88%. At \$50K notional the executable rate drops to 1.62%; at \$500K, virtually no windows are executable (0.03%). The conclusion is qualitatively identical across all regimes.

Meta-labeling null result. The MetaLabelingFilter is trained on windows where the primary price signal fires ($|b_{\text{USDC}}| >$ 10 bps). In the train split only 53 such windows exist, and zero are net-profitable consistent with the zero positive rate in calm-control training data (Table 1). The meta-model learns no positive class signal and takes 0 trades on the

Table 8: Price-to-execution ratio across cost regimes ($10K, 10 bps, 5-min horizon, test split).

Settlement	Fee regime	Exec. %	Ratio
0 bps	low (institutional)	6.11%	2.07×
0 bps	base	5.64%	2.24×
0 bps	high	5.46%	2.32×
5 bps	base	5.16%	2.45×
10 bps	base	4.88%	2.59×
10 bps	high (worst case)	4.81%	2.63×

Table 9: Regime detection (test split, stress = $|b_{\mathrm{USDC}}| >$ 10 bps).

Detector	Acc.	Recall	FAR	AUROC
EWMA (span=20, thr=2.5)	0.869	0.010	0.007	0.624
CUSUM (k=0.5, h=5)	0.270	1.000	0.835	0.582
BOCPD (hz=0.01, thr=0.5)	0.860	0.022	0.019	0.229

FAR = false alarm rate.

test split. This is an informative null: meta-labeling is architecturally sound but requires a training regime with both positive and negative meta-labels. It motivates using the Terra/LUNA validation split for meta-model training in future work.

Regime detectors. Table 9 shows the recall-precision tradeoff. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% false alarm rate with very low recall (0.95%); BOCPD achieves only AUROC 0.229 on the CEX cross-quote series. No detector alone solves the execution identification problem; their value is as a pre-filter that reduces the search space.

False-positive diagnosis. Applying `PriceBasis10bps` to the test split produces 2,002 trades. False positives 581 windows where basis > 10 bps but order-book depth is insufficient account for 29% of trades at a mean net loss of −38.7 bps. This directly illustrates the execution-barrier problem: the rule fires on price dislocations where the book is too thin to fill the arbitrage leg.

6.5 Stress Recurs Across Mechanisms, But Execution Labels Are Scarce

The benchmark is evaluated on a single Tier-A stress event, which raises the question of whether the price-to-execution gap and oracle gap are specific to the USDC/SVB episode. The historical catalogue (Table 3) provides indirect evidence that execution barriers are a structural property of stablecoin stress rather than an artefact of one event.

Across seven mechanism classes, stress events share common features: a price dislocation arises rapidly, order-book depth deteriorates at the moment when arbitrage would be most profitable, and settlement risk rises simultaneously. This pattern holds regardless of whether the stress originates from a bank-reserve shock (USDC/SVB), an algorithmic-reflexive collapse (Terra/UST), an exchange credit event (FTX), a DeFi pool imbalance (USDT/Curve), or a regulatory wind-down (BUSD).

The constraint is not whether stress occurs it does, repeatedly but whether execution-grade order-book data exists to measure the gap precisely. Only Tier-A events provide the L2 depth required for VWAP label construction. The Terra/LUNA validation split (N=11,523 minutes, 13.76% price-basis positive, 3.08% executable) demonstrates similar dynamics under an independent mechanism, but without the full Tier-A execution reconstruction needed for P&L claims. Expanding Tier-A coverage to additional mechanism classes is the primary path to testing whether the oracle gap generalises across stress regimes.

7 LIMITATIONS

Tier-A execution-grade history is scarce. The test split covers one execution-grade stress event (SVB March 2023). Future work should extend Tier-A coverage to additional mechanism classes; the 18-event catalogue documents the data requirements. Models trained on fiat-reserve bank shocks may not generalise to algorithmic-reflexive or exchange-credit failure modes.

Historical catalogue is broader than benchmark. Tier-B and Tier-C events support price and taxonomy claims only. The benchmark deliberately separates what can be claimed from Tier-A data from what can only be described at Tier-B or Tier-C.

Settlement and latency are proxied. CEX-to-CEX settlement latency is approximated via a fixed penalty parameter; true on-chain settlement delay and block confirmation times are not modelled. A more detailed treatment would require per-block latency data beyond the scope of this benchmark.

Venue-depth coverage. Binance USDM futures bookDepth is the confirmed execution-grade source in the committed dataset. Full historical L2 for Coinbase and Kraken during the SVB period requires a Tardis subscription not included in the public release.

Model scope. The ladder covers tabular ML and statistics-based approaches. Deep sequence models, graph networks, and RL-based order-execution policies are outside scope and represent natural extensions. The oracle gap defines the challenge they must close.

8 CONCLUSION

Stablecoins promise price stability, but stress events create visible dislocations across venues. Those dislocations look like arbitrage on a chart. The central result of this paper is that they mostly are not: 35.1% of minutes in the USDC/SVB stress window show a cross-quote basis above 10 bps, but only 2.88% remain profitable after VWAP book-walking and costs a 12× price-to-execution gap.

Profitable windows do exist: a hindsight oracle earns +161.7 bps per trade, confirming that execution-grade opportunities are real, not artefacts of the label construction. Every active non-oracle strategy that takes trades price rules, ML classifiers, tree ensembles, and a direct expected-net-profit regressor produces negative net bps out of sample; abstention remains the zero-P&L floor. The oracle gap exceeds 210 bps across all tasks and persists across fee regimes, settlement penalties, and notional sizes.

Stablecoin StressBench shows that the main challenge is not observing price dislocations, but predicting which dislocations survive execution. The benchmark defines an oracle gap between hindsight-executable opportunities and ex-ante deployable models. The historical catalogue shows why this benchmark matters beyond one event: stablecoin stress recurs across mechanisms, while only some episodes provide the depth required for execution-aware evaluation. Closing the oracle gap through better microstructure features, improved calibration, or deeper sequence and network models is the central challenge for future work on crypto market microstructure, financial ML validation, and execution-aware trading systems.

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REFERENCES

[1] Kwon, S., Minegishi, K., and Nishi, R. (2023). Terra/LUNA depeg mechanics: Algorithmic stablecoin collapse and contagion dynamics. Working Paper.
[2] Griffin, J. M. and Shams, A. (2020). Is Bitcoin really untethered? Journal of Finance, 75(4):1913–1964.
[3] Hautsch, N., Scheuch, C., and Voigt, S. (2018). Limits to arbitrage in markets with stochastic settlement latency. Working Paper, Vienna Graduate School of Finance.
[4] Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. Journal of Finance, 52(1):35–55.
[5] Gromb, D. and Vayanos, D. (2010). Limits of arbitrage: The state of the theory. Annual Review of Financial Economics, 2:251–275.
[6] Cont, R. and Kukanov, A. (2013). Optimal order placement in limit order markets. Quantitative Finance, 17(1):21–39.
[7] Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics, 14(1):71–100.
[8] Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. Journal of Financial Economics, 135(2):293–319.
[9] Adams, H., et al. (2021). Uniswap v3 core. Technical Report, Uniswap Labs.
[10] López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley. [Meta-labeling framework: Chapter 3.]
[11] Cintra, R. and Holloway, C. (2023). Bayesian online changepoint detection for stablecoin depeg events: Evidence from Curve StableSwap pools. Working Paper.
[12] Vidal-Tomàs, D., Briola, A., and Aste, T. (2023). FTX's downfall and Binance's consolidation: The fragility of centralised exchanges. SSRN Working Paper.